BIOCHEMISTRY CUMULATIVE EXAM

January 15, 1994

In many cases involving electron transfer between proteins, electron transfer can occur only after formation of a precursor complex between the two proteins involved.

$$A_{ox} + B_{red} \underset{K_d}{\Longleftrightarrow} A_{ox}:B_{red} \xrightarrow{k_{et}} A_{red}:B_{ox} \longrightarrow A_{red} + B_{ox}$$

The rate observed rate constant, k_{obs}, is related to k_{et}, the first-order rate constant for electron transfer within the complex, and K_d, the dissociation constant for the complex by the following equation

$$k_{obs} = k_{et} [A]/(K_d + [A])$$

under conditions where [B] is constant.

a. (15 points) For the electron transfer reaction between native A and B, K_d and k_{et} have been experimentally determined to be 2.5 μM and 1000 sec^{-1}, respectively. What is the expected k_{obs} when [A] = 100 μM?

b. (15 points) Alterations in the structure of A by either chemical modification or site selective mutagenesis may be useful in assessing the roles of particular amino acid residues in the interaction between the two proteins and/or in the electron transfer process itself. If a change in a residue affected only the interaction between the two proteins, what effect would you expect it to have on K_d and k_{et}? If a change in a residue affected only the electron transfer between the two proteins, what effect would you expect it to have on K_d and k_{et}?

c. (50 points) Let us assume that we have used site directed mutagenesis to alter two sites on protein A. The changes made involve two neighboring acidic residues Asp60 and Asp61. In each mutated protein the Asp has been changed to a Lys; designated as Asp60Lys or Asp61Lys. In independent binding studies between B_{ox} and the native and mutated A_{ox}, the K_d values have been obtained. The k_{obs} has also been determined for the reaction between A_{ox} (= 100 μM) and B_{red} (constant) in each case.

	K_d from independent binding study	k_{obs} at [A] = 100 μM
A (native)	2.5 μM	see a. above
A Asp60Lys	25.0 μM	800 sec^{-1}
A Asp61Lys	25.0 μM	400 sec^{-1}

Based on this information what would you conclude regarding the roles of the indicated amino acid residues in (i) the interaction between the two proteins and (ii) in the electron transfer process? Justify your answers carefully.

d. (20 points) In c., did you make any assumptions which might be questionable? If so, what were they? How might you experimentally address the validity of such assumptions?